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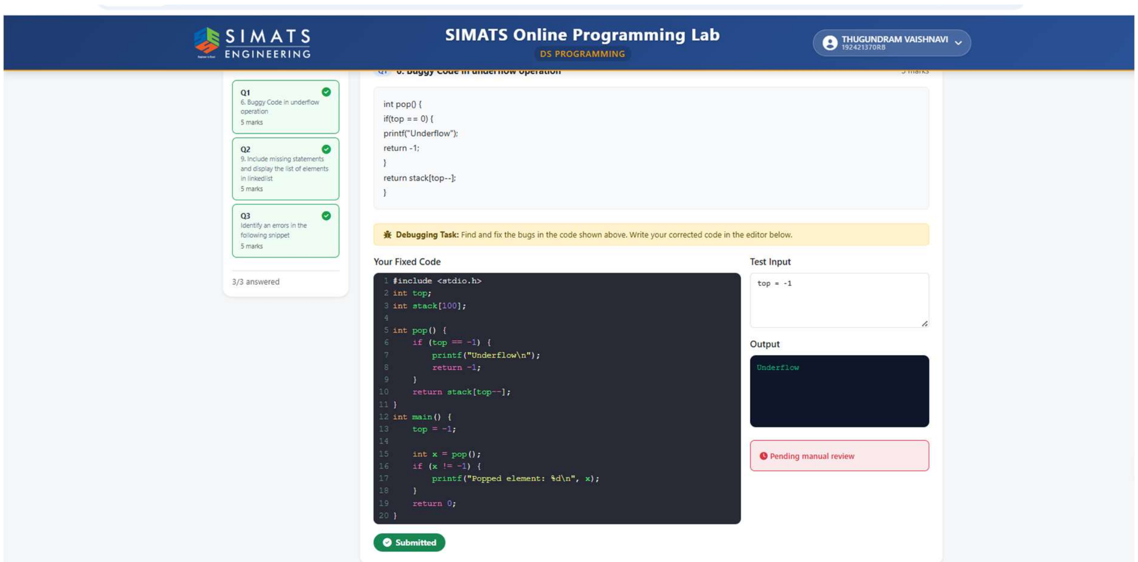
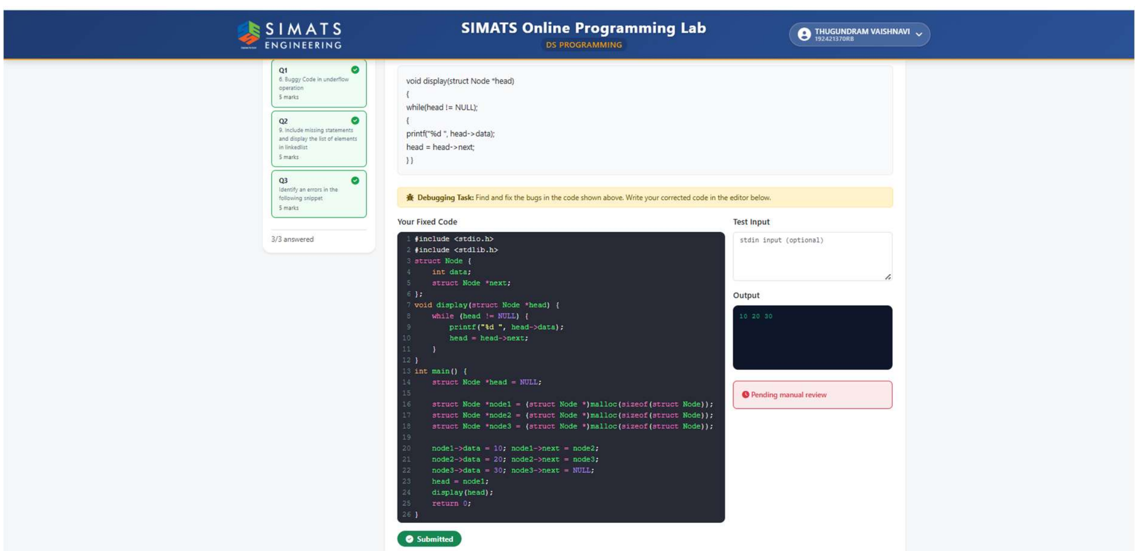
DEBUGGING -CO2

CSA0312-DATASTRUCTURE





QUESTIONS

Q1
6. Buggy Code in underflow operation
5 marks

Q2
9. Include missing statements and display the list of elements in linkedlist
5 marks

Q3
Identify an errors in the following snippet
5 marks

3/3 answered

Q3 Identify an errors in the following snippet

5 marks

```
int binarySearch(int arr[], int n, int key)
{
int low = 0, high = n - 1;
while(low < high)
{ int mid = (low + high) / 2;
if(arr[mid] == key) return mid;
else if(arr[mid] < key)
low = mid;
else high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;
    while (low <= high)
    {
        int mid = low + (high - low) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}
int main()
{
    int arr[] = {1, 3, 5, 7, 9};
    int n = 5;
    int key = 7;
    int ans = binarySearch(arr, n, key);
    printf("%d\n", ans);
    return 0;
}
```

Test Input

```
arr = [1,3,5,7,9]
n = 5
key = 7
```

Output

```
3
```

Pending manual review

Submitted